

22 Jan 2026

Periodic orbits in the plane

Bendixon's Theorem:

Time-invariant planar system

$$\dot{x}_1 = f_1(x_1, x_2)$$

$$\nabla f(x) = \frac{\partial f_1}{\partial x_1} + \frac{\partial f_2}{\partial x_2}$$

$$\dot{x}_2 = f_2(x_1, x_2)$$

if $\nabla f(x)$ is not identically zero

and does not change sign

in a simply connected region D

then there are no periodic orbits exist that lie entirely in D



Simply connected but not convex:

for a linear system,

$$f(x_1, x_2) = A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = A$$

$$\nabla f(x) = \frac{\partial f_1}{\partial x_1} + \frac{\partial f_2}{\partial x_2} = \text{trace}(A)$$

For a periodic orbit to exist in a linear system, it is necessary for $\text{trace}(A) = 0$

Periodic orbit $\dot{x} = \begin{bmatrix} 0 & -\beta \\ \beta & 0 \end{bmatrix} x$

$$\dot{x} = \begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix} x \quad \alpha \neq 0$$

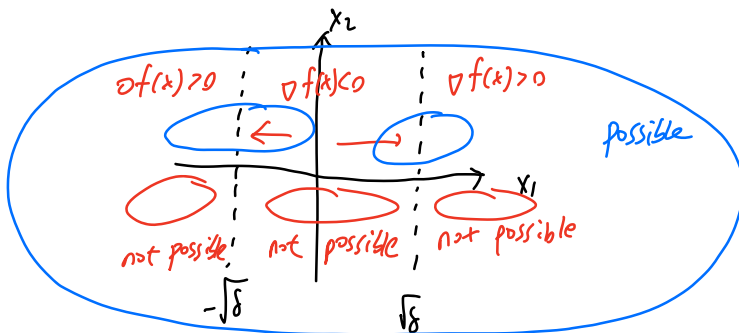
$$\text{trace}(A) = 2\alpha$$

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x \quad \text{tr}(A) = 0$$

Ex. $\dot{x}_1 = x_2$ $\delta > 0$

$$\dot{x}_2 = -\delta x_2 + x_1 - x_1^3 + x_1^2 x_2$$

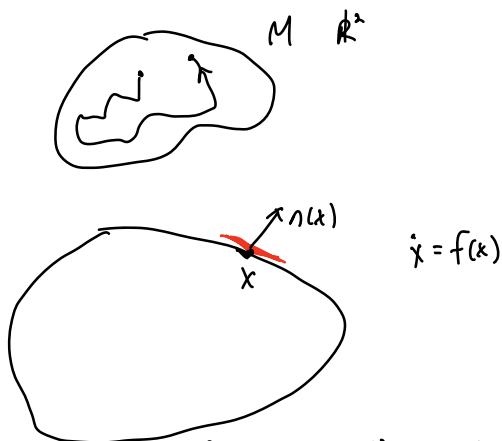
$$\begin{aligned} \nabla f(x) &= \frac{\partial f}{\partial x_1} + \frac{\partial f}{\partial x_2} \\ &= 0 + (-\delta + x_1^2) \\ &= x_1^2 - \delta \end{aligned}$$



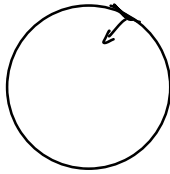
Poincaré - Bendixson Theorem

Invariant set

"A set $M \subset \mathbb{R}^2$ is (positively) invariant, if for each initial condition in M , the trajectory remains in M $\forall t \geq 0$ "



$f(x) \cdot n(x) \leq 0 \quad \forall x \text{ on the boundary of } M \Rightarrow M \text{ is positively invariant}$



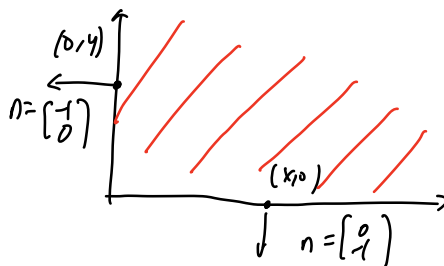
$f(x) \cdot n(x) < 0 \quad \forall x$ on the boundary of this circle
 $\Rightarrow$ circle is positively invariant

under-approximation of the maximal (pb) invariant set

Ex. Predator-Prey Model $a, b, c, d > 0$

$$\dot{x} = (a - bx) x \quad \text{Prey (exponential growth when } y=0)$$

$$\dot{y} = (cx - d) y$$



$$f \cdot n \leq 0$$

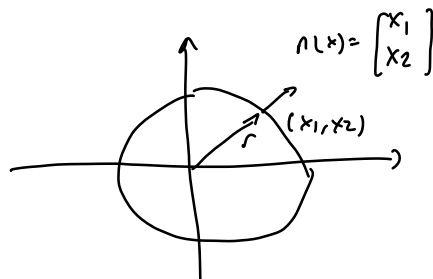
$$\Rightarrow (cx - d) \cdot y \cdot (-1) \leq 0 \quad \text{for } y=0$$

$$= 0$$

Ex. $\dot{x}_1 = x_1 + x_2 - x_1(x_1^2 + x_2^2)$

$$\dot{x}_2 = -2x_1 + x_2 - x_2(x_1^2 + x_2^2)$$

$\{x: x_1^2 + x_2^2 \leq r\}$ is positively invariant for sufficiently large r



$$f(x) \cdot n(x) = x_1^2 + x_1 x_2 - x_1^2(x_1^2 + x_2^2)$$

$$- 2x_1 x_2 + x_2^2 - x_2^2(x_1^2 + x_2^2)$$

$$= x_1^2 + x_2^2 - (x_1^2 + x_2^2)^2 - x_1 x_2$$

$$= r^2 - r^4 - x_1 x_2$$

$$(x_1 + x_2)^2 \geq 0$$

$$x_1^2 + x_2^2 + 2x_1 x_2 \geq 0$$

$$-x_1 x_2 \leq \frac{x_1^2 + x_2^2}{2}$$

$$-x_1 x_2 \leq \frac{r^2}{2}$$

$$f(x) \cdot \eta(x) = r^2 - r^4 - x_1 x_2$$

$$\leq r^2 - r^4 + \frac{r^2}{2}$$

$$= \frac{3}{2} r^2 - r^4$$

$$= r^2 \left(\frac{3}{2} - r^2 \right)$$

$$\leq 0 \quad \text{if } r \geq \sqrt{\frac{3}{2}}$$

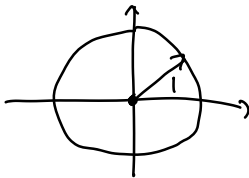
Poincaré-Bendixon Theorem:

Suppose M is compact (closed and bounded), and positively invariant

for a planar time invariant system $\dot{x} = f(x)$, $x \in \mathbb{R}^2$

If M contains no equilibrium points, then it contains a periodic orbit

Ex. $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad \begin{aligned} \dot{x}_1 &= -x_2 \\ \dot{x}_2 &= x_1 \end{aligned}$



M is compact
and positively invariant
and does not include eq.
 $\Rightarrow \exists$ a periodic orbit in M
Poincaré-Bendixon

Bifurcations

A bifurcation is an abrupt change in qualitative behavior as a parameter is varied.

Example: equilibria or limit cycles appearing/disappearing becoming stable/unstable



• Transition: multiple equilibria $\longleftrightarrow$ no equilibrium

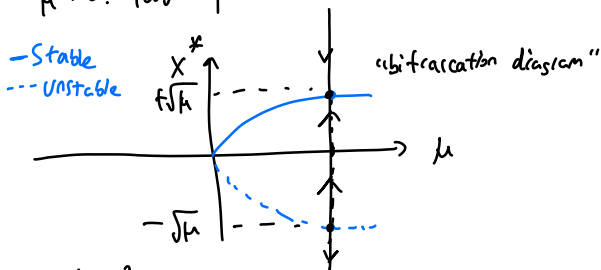
Fold bifurcation, "blue-sky" bifurcation, or "saddle node" bifurcation

$$\dot{x} = \underbrace{\mu - x^2}_{f(x)} \quad f(x)=0 \text{ has solution only if } \mu \geq 0$$

$\mu < 0$: no equilibrium

$\mu = 0$: one equilibrium

$\mu > 0$: two equilibria



$$\dot{x} = \mu - x^2$$

$$\dot{\tilde{x}} = \frac{\partial f}{\partial x} \bigg|_{x=x^*} (\tilde{x} - x^*)$$

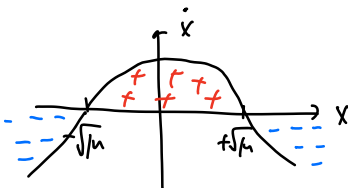
$$\dot{x} = \sqrt{\mu}$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (\mu - x^2)$$

$$= -2x \bigg|_{x=\sqrt{\mu}}$$

$$= -2\sqrt{\mu}$$

$$\dot{\tilde{x}} = -2\sqrt{\mu} (\tilde{x} - \sqrt{\mu})$$



• Transcritical bifurcation

$$\dot{x} = \underbrace{\mu x - x^2}_{f(x)} \quad \mu x - x^2 = 0$$

$$\Rightarrow x = 0 \text{ and } x = \mu$$

$\mu < 0$: $x = 0$ is stable

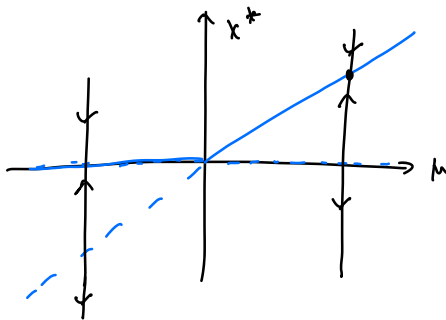
$x = \mu$ is unstable

$\mu > 0$: $x = 0$ is unstable

$x = \mu$ is stable

$$\frac{\partial f}{\partial x} \Big|_{x=0} = \mu - 2x \Big|_{x=0} = \mu$$

$$\frac{\partial f}{\partial x} \Big|_{x=\mu} = \mu - 2x \Big|_{x=\mu} = -\mu$$



Pitchfork bifurcation

$$\dot{x} = \mu x - x^3$$

$$\mu x - x^3 = 0$$

$\mu < 0$: $x = 0$ always

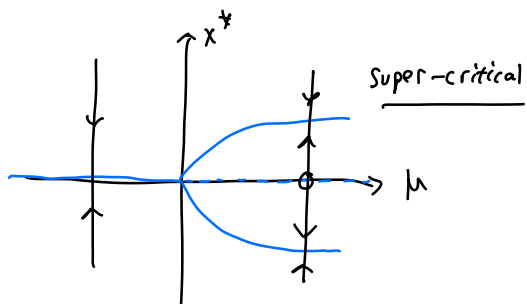
$\mu > 0$: $x = 0$, $x = \pm\sqrt{\mu}$

	$\mu < 0$	$\mu > 0$
$x = 0$	stable	unstable
$x = +\sqrt{\mu}$	N/A	stable
$x = -\sqrt{\mu}$	N/A	stable

$$\frac{\partial f}{\partial x} \Big|_{x=0} = \mu - 3x^2 \Big|_{x=0} = \mu$$

$$\frac{\partial f}{\partial x} \Big|_{x=\sqrt{\mu}} = \mu - 3x^2 \Big|_{x=\sqrt{\mu}} = -2\mu$$

$$\frac{\partial f}{\partial x} \Big|_{x=-\sqrt{\mu}} = \mu - 3x^2 \Big|_{x=-\sqrt{\mu}} = -2\mu$$



$$\dot{x} = \mu x + x^3$$

